

```

+ tags_all = {
+   "Name" = "mi-main"
+ }
}

# aws_vpc_security_group_egress_rule.all will be created
+ resource "aws_vpc_security_group_egress_rule" "all" {
+   arn = (known after apply)
+   cidr_ipv4 = "0.0.0.0/0"
+   id = (known after apply)
+   ip_protocol = "-1"
+   security_group_id = (known after apply)
+   security_group_rule_id = (known after apply)
+   tags_all = {}
+ }

# aws_vpc_security_group_ingress_rule.ssh will be created
+ resource "aws_vpc_security_group_ingress_rule" "ssh" {
+   arn = (known after apply)
+   cidr_ipv4 = "0.0.0.0/0"
+   from_port = 22
+   id = (known after apply)
+   ip_protocol = "tcp"
+   security_group_id = (known after apply)
+   security_group_rule_id = (known after apply)
+   tags_all = {}
+   to_port = 22
+ }

```

Plan: 19 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

Admin@DESKTOP-U0T8F2T MINGW64 /d/danit_homework/HW19/terraform (main)

\$

```

}

```

Plan: 19 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

```

aws_eip.nat: Creating...
aws_vpc.main: Creating...
aws_eip.nat: Creation complete after 1s [id=eipalloc-09a2fa475377afb54]
aws_vpc.main: Creation complete after 1s [id=vpc-067c90891adf9ec86]
aws_internet_gateway.main: Creating...
aws_security_group.main: Creating...
aws_subnet.public_a: Creating...
aws_subnet.public_b: Creating...
aws_subnet.private_a: Creating...
aws_subnet.private_b: Creating...
aws_internet_gateway.main: Creation complete after 1s [id=igw-0b3dc090668db3875]
aws_route_table.public: Creating...
aws_subnet.public_a: Creation complete after 1s [id=subnet-009e50a38e68d3338]
aws_nat_gateway.main: Creating...
aws_route_table.public: Creation complete after 1s [id=rtb-056b0a54ee4b3ac39]
aws_route_table_association.public_a: Creating...
aws_route_table_association.public_a: Creation complete after 0s [id=rtbassoc-0e4f9ba6dc38383]
aws_security_group.main: Creation complete after 3s [id=sg-0eb407bfb3e64fc90]
aws_instance.ec2public: Creating...
aws_vpc_security_group_ingress_rule.ssh: Creating...
aws_vpc_security_group_egress_rule.all: Creating...
aws_subnet.private_b: Creation complete after 2s [id=subnet-06657b11c07ed613d]
aws_vpc_security_group_ingress_rule.ssh: Creation complete after 0s [id=sgr-0bb7ef9ba6dc38383]
aws_vpc_security_group_egress_rule.all: Creation complete after 0s [id=sgr-0d3ad0fe495e3152f]
aws_subnet.public_b: Creation complete after 4s [id=subnet-0e4a38d78f430bfd2]
aws_route_table_association.public_b: Creating...
aws_subnet.private_a: Creation complete after 4s [id=subnet-0cbbd5ce53ab143b3]
aws_instance.ec2private: Creating...
aws_route_table_association.public_b: Creation complete after 1s [id=rtbassoc-076ca8b2c9afd54d1]
aws_nat_gateway.main: Still creating... [00m11s elapsed]

```

```
+ cpu_options (known after apply)
+ ebs_block_device (known after apply)
+ enclave_options (known after apply)
+ ephemeral_block_device (known after apply)
+ instance_market_options (known after apply)
+ maintenance_options (known after apply)
+ metadata_options (known after apply)
+ network_interface (known after apply)
+ private_dns_name_options (known after apply)
+ root_block_device (known after apply)
}
```

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```
aws_instance.ec2private: Creating...
aws_instance.ec2public: Creating...
aws_instance.ec2private: Still creating... [00m10s elapsed]
aws_instance.ec2public: Still creating... [00m10s elapsed]
aws_instance.ec2public: Creation complete after 13s [id=i-05a0fe79a6918cbc2]
aws_instance.ec2private: Creation complete after 13s [id=i-051e522b0bbe22464]
```

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Admin@DESKTOP-U0T8F2T MINGW64 /d/danit_homework/HW19/terraform (main)

The screenshot shows the Amazon S3 console interface. On the left, there's a navigation pane with 'Amazon S3' selected. The main area shows the 'network/' bucket. Under the 'Objects' tab, a table lists the objects in the bucket. There is one object named 'terraform.tfstate' with a size of 36.8 KB and a storage class of 'Standard'. The object was last modified on July 28, 2026, at 18:37:53 (UTC+03:00).

The screenshot shows the AWS VPC console interface. The left navigation pane has 'VPC dashboard' selected. The main area shows the details of a VPC named 'vpc-067c90891adf9ec86 / mi-main'. The 'Details' tab is selected, displaying various configuration options. The VPC ID is 'vpc-067c90891adf9ec86', the state is 'Available', and DNS resolution is 'Enabled'. The main network ACL is 'acl-04ecc0598f4f02ad3'. The IP address pool is '10.0.0.0/16'. The route table is 'rtb-0909d55e31666b543'. The VPC is associated with the 'mi-main' subnet. The console also shows a 'Resource map' at the bottom, which visualizes the VPC's components, including subnets, route tables, and network connections.

igw-0b3dc090668db3875 / mi-igw

Details

Internet gateway ID: [igw-0b3dc090668db3875](#) State: Attached VPC ID: [vpc-067c90891ad9ecb6](#) / [mi-main](#) Owner: [324621154857](#)

Tags (1)

Key	Value
Name	mi-igw

3.68.59.90

Summary

Allocated IPv4 address: [3.68.59.90](#) Type: [Public IP](#) Allocation ID: [eipalloc-09a2f6475377afb54](#) Reverse DNS record: [-](#)

Association ID: [eipassoc-05dfe11bd7cfd5a5c](#) Scope: [VPC](#) Associated Instance ID: [-](#) Private IP address: [10.0.1.122](#)

Network interface ID: [eni-0a15a2b30274d7408](#) Network interface owner account ID: [324621154857](#) Public DNS: [-](#) NAT Gateway ID: [nat-0f605513d0ab1d5b4](#) / [mi-nat](#)

Address pool: [Amazon](#) Network border group: [eu-central-1](#) Service managed: [-](#)

Tags (1)

Key	Value
Name	mi-nat

nat-0f605513d0ab1d5b4 / mi-nat

Details

NAT gateway ID: [nat-0f605513d0ab1d5b4](#) Connectivity type: [Public](#) State: Available State message: [-](#)

NAT gateway ARN: [arn:aws:ec2:eu-central-1:324621154857:natgateway/nat-0f605513d0ab1d5b4](#) Primary public IPv4 address: [3.68.59.90](#) Primary private IPv4 address: [10.0.1.122](#) Primary network interface ID: [eni-0c1563302746c508](#)

VPC: [vpc-067c90891ad9ecb6](#) / [mi-main](#) Subnet: [subnet-009c50a38e68d338](#) / [mi-public-a](#) Created: [Tuesday, July 28, 2026 at 18:20:45 GMT+3](#) Deleted: [-](#)

Secondary IPv4 addresses

Secondary IPv4 addresses are not available for this nat gateway.

Instances (2)

Find instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security g
mi-ec2private	i-051e52250bba22464	Running	t3.micro	3/3 checks pass	eu-central-1a	-	-	-	-	disabled	mi-sg
mi-ec2public	i-05a0fc79a6918dc2	Running	t3.micro	3/3 checks pass	eu-central-1a	-	3.67.64.158	-	-	disabled	mi-sg

sg-0eb407bfb3e64fc90 - mi-sg

Details

Security group name: [mi-sg](#) Security group ID: [sg-0eb407bfb3e64fc90](#) Description: [TF security group](#) VPC ID: [vpc-067c90891ad9ecb6](#)

Owner: [324621154857](#) Inbound rules count: [1](#) Outbound rules count: [1](#)

Inbound rules (1)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-0bb7ef9ba6dc38383	IPv4	SSH	TCP	22	0.0.0.0/0	-

EC2 > Instances

Instances (1/2) info

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security g
mi-ec2-private	i-051e522b0bbe22464	Running	t3.micro	3/3 checks passed	eu-central-1a	-	-	-	-	disabled	mi-sg
mi-ec2-public	i-05a0fe79a69181c62	Running	t3.micro	3/3 checks passed	eu-central-1a	-	3.67.64.158	-	-	disabled	mi-sg

i-051e522b0bbe22464 (mi-ec2-private)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary

Instance ID i-051e522b0bbe22464	Public IPv4 address -	Private IPv4 addresses 10.0.3.19
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-0-3-19-eu-central-1-compute.internal	Private IP DNS name (IPv4 only) ip-10-0-3-19-eu-central-1-compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address -	VPC ID vpc-067c90891aaf9ec86 (mi-main)	Auto Scaling Group name -
IAM role -	Subnet ID -	

```
Admin@DESKTOP-U0T8F2T MINGW64 /d/danit_homework/HW19/terraform (main)
$ ssh -i my-key.pem ubuntu@3.67.64.158
The authenticity of host '3.67.64.158 (3.67.64.158)' can't be established.
ED25519 key fingerprint is: SHA256:2BlykzN7Uobl.duyISUTUuej1brvwTG67EUXdIE6D5aTU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.67.64.158' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1061-aws x86_64)
```

```
* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:      https://ubuntu.com/pro
```

System information as of Tue Jul 28 16:04:40 UTC 2026

```
System load: 0.0          Processes:            104
Usage of /: 23.0% of 7.57GB Users logged in:      0
Memory usage: 25%        IPv4 address for ens5: 10.0.1.77
Swap usage: 0%
```

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

```
ubuntu@ip-10-0-1-77:~$
ubuntu@ip-10-0-1-77:~$
```

```

Admin@DESKTOP-U0T8F2T MINGW64 /d/danit_homework/HW19/terraform (main)
• $ eval "$(ssh-agent -s)"
  Agent pid 2030

Admin@DESKTOP-U0T8F2T MINGW64 /d/danit_homework/HW19/terraform (main)
• $ ssh-add ~/my-key.pem
  /c/Users/Admin/my-key.pem: No such file or directory

Admin@DESKTOP-U0T8F2T MINGW64 /d/danit_homework/HW19/terraform (main)
• $ ssh-add my-key.pem
  Identity added: my-key.pem (my-key.pem)

Admin@DESKTOP-U0T8F2T MINGW64 /d/danit_homework/HW19/terraform (main)
• $ ssh -A -J ubuntu@3.67.64.158 ubuntu@10.0.3.19
  The authenticity of host '10.0.3.19 (<no hostip for proxy command>)' can't be established.
  ED25519 key fingerprint is: SHA256:UR0c7A+v0xtzgAGXx1b0fgnzaSYynHq0YE0D7y40zRw
  This key is not known by any other names.
  Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
  Warning: Permanently added '10.0.3.19' (ED25519) to the list of known hosts.
  Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1061-aws x86_64)

  * Documentation:  https://help.ubuntu.com
  * Management:    https://landscape.canonical.com
  * Support:       https://ubuntu.com/pro

System information as of Tue Jul 28 16:10:10 UTC 2026

System load:  0.0          Processes:      103
Usage of /:   22.9% of 7.57GB Users logged in:  0
Memory usage: 23%         IPv4 address for ens5: 10.0.3.19
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

```

```

Admin@DESKTOP-U0T8F2T MINGW64 /d/danit_homework/HW19/terraform (main)
$ ssh -A -J ubuntu@3.67.64.158 ubuntu@10.0.3.19
The authenticity of host '10.0.3.19 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is: SHA256:UR0c7A+v0xtzgAGXx1b0fgnzaSYynHq0YE0D7y40zRw
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.3.19' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1061-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Tue Jul 28 16:10:10 UTC 2026

System load:  0.0          Processes:      103
Usage of /:   22.9% of 7.57GB Users logged in:  0
Memory usage: 23%         IPv4 address for ens5: 10.0.3.19
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-10-0-3-19:~$

```

```

ubuntu@ip-10-0-3-19:~$ ping google.com
PING google.com (142.251.14.100) 56(84) bytes of data.
64 bytes from pm-in-f100.1e100.net (142.251.14.100): icmp_seq=1 ttl=113 time=2.70 ms

64 bytes from pm-in-f100.1e100.net (142.251.14.100): icmp_seq=2 ttl=113 time=1.32 ms
64 bytes from pm-in-f100.1e100.net (142.251.14.100): icmp_seq=3 ttl=113 time=1.33 ms
64 bytes from pm-in-f100.1e100.net (142.251.14.100): icmp_seq=4 ttl=113 time=1.31 ms
64 bytes from pm-in-f100.1e100.net (142.251.14.100): icmp_seq=5 ttl=113 time=1.32 ms
64 bytes from pm-in-f100.1e100.net (142.251.14.100): icmp_seq=6 ttl=113 time=1.35 ms
64 bytes from pm-in-f100.1e100.net (142.251.14.100): icmp_seq=7 ttl=113 time=1.32 ms
^C
--- google.com ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6010ms
rtt min/avg/max/mdev = 1.311/1.522/2.699/0.480 ms
ubuntu@ip-10-0-3-19:~$

```

```

- owner_id              = "324621154857" -> null
- tags                  = {
  - "Name" = "mi-main"
} -> null
- tags_all              = {
  - "Name" = "mi-main"
} -> null
# (4 unchanged attributes hidden)
}

# aws_vpc_security_group_egress_rule.all will be destroyed
- resource "aws_vpc_security_group_egress_rule" "all" {
  - arn              = "arn:aws:ec2:eu-central-1:324621154857:security-group-rule/sgr-0d3ad0fe495e3152f" -> null
  - cidr_ipv4        = "0.0.0.0/0" -> null
  - id               = "sgr-0d3ad0fe495e3152f" -> null
  - ip_protocol       = "-1" -> null
  - security_group_id = "sg-0eb407bfb3e64fc90" -> null
  - security_group_rule_id = "sgr-0d3ad0fe495e3152f" -> null
  - tags_all         = {} -> null
}

# aws_vpc_security_group_ingress_rule.ssh will be destroyed
- resource "aws_vpc_security_group_ingress_rule" "ssh" {
  - arn              = "arn:aws:ec2:eu-central-1:324621154857:security-group-rule/sgr-0bb7ef9ba6dc38383" -> null
  - cidr_ipv4        = "0.0.0.0/0" -> null
  - from_port        = 22 -> null
  - id               = "sgr-0bb7ef9ba6dc38383" -> null
  - ip_protocol       = "tcp" -> null
  - security_group_id = "sg-0eb407bfb3e64fc90" -> null
  - security_group_rule_id = "sgr-0bb7ef9ba6dc38383" -> null
  - tags_all         = {} -> null
  - to_port           = 22 -> null
}

```

Plan: 0 to add, 0 to change, 19 to destroy.

Do you really want to destroy all resources?
 Terraform will destroy all your managed infrastructure, as shown above.
 There is no undo. Only 'yes' will be accepted to confirm.

Enter a value:

```

aws_subnet.public_b: Destroying... [id=subnet-0e4a38d78f430bfd2]
aws_route_table.public: Destroying... [id=rtb-056b0a54ee4b3ac39]
aws_subnet.private_b: Destroying... [id=subnet-06657b11c07ed613d]
aws_route_table.private: Destroying... [id=rtb-0aeefe55fb54448df]
aws_vpc_security_group_egress_rule.all: Destruction complete after 0s
aws_subnet.public_b: Destruction complete after 1s
aws_route_table.public: Destruction complete after 1s
aws_subnet.private_b: Destruction complete after 1s
aws_route_table.private: Destruction complete after 1s
aws_nat_gateway.main: Destroying... [id=nat-0f605513d0ab1d5b4]
aws_instance.ec2public: Still destroying... [id=i-05a0fe79a6918cbc2, 00m10s elapsed]
aws_instance.ec2private: Still destroying... [id=i-051e522b0bbe22464, 00m10s elapsed]
aws_nat_gateway.main: Still destroying... [id=nat-0f605513d0ab1d5b4, 00m10s elapsed]
aws_instance.ec2public: Still destroying... [id=i-05a0fe79a6918cbc2, 00m20s elapsed]
aws_instance.ec2private: Still destroying... [id=i-051e522b0bbe22464, 00m20s elapsed]
aws_nat_gateway.main: Still destroying... [id=nat-0f605513d0ab1d5b4, 00m20s elapsed]
aws_instance.ec2public: Still destroying... [id=i-05a0fe79a6918cbc2, 00m30s elapsed]
aws_instance.ec2private: Still destroying... [id=i-051e522b0bbe22464, 00m30s elapsed]
aws_instance.ec2private: Destruction complete after 30s
aws_subnet.private_a: Destroying... [id=subnet-0cbbd5ce53ab143b3]
aws_nat_gateway.main: Still destroying... [id=nat-0f605513d0ab1d5b4, 00m30s elapsed]
aws_subnet.private_a: Destruction complete after 0s
aws_instance.ec2public: Still destroying... [id=i-05a0fe79a6918cbc2, 00m40s elapsed]
aws_instance.ec2public: Destruction complete after 40s
aws_security_group.main: Destroying... [id=sg-0eb407bfb3e64fc90]
aws_nat_gateway.main: Still destroying... [id=nat-0f605513d0ab1d5b4, 00m40s elapsed]
aws_security_group.main: Destruction complete after 1s
aws_nat_gateway.main: Destruction complete after 40s
aws_internet_gateway.main: Destroying... [id=igw-0b3dc090668db3875]
aws_subnet.public_a: Destroying... [id=subnet-009e50a38e68d3338]
aws_eip.nat: Destroying... [id=eipalloc-09a2fa475377afb54]
aws_internet_gateway.main: Destruction complete after 1s
aws_subnet.public_a: Destruction complete after 1s
aws_vpc.main: Destroying... [id=vpc-067c90891adf9ec86]
aws_eip.nat: Destruction complete after 1s
aws_vpc.main: Destruction complete after 1s

```

Destroy complete! Resources: 19 destroyed.

Admin@DESKTOP-U0T8F2T MINGW64 /d/danit_homework/HW19/terraform (main)

The screenshot shows the AWS Management Console interface. On the left is a sidebar with navigation links for 'Virtual private cloud' resources. The main content area is titled 'Your VPCs' and contains a table listing the VPCs in the account. A single VPC is listed with the ID 'vpc-053412dc3f73c0ea' and a status of 'Available'. The table headers include Name, VPC ID, State, Encryption c..., Encryption control..., Block Public..., IPv4 CIDR, IPv6 CIDR, DHCP option set, and Main rou. The console also features a top navigation bar with the AWS logo, a search bar, and a 'Your VPCs' breadcrumb.

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set	Main rou
-	vpc-053412dc3f73c0ea	Available	-	-	Off	172.31.0.0/16	-	default-0ed66e0f59777c43	rtb-0c2b...